

Project Design Phase

Solution Architecture

Date	15 February 2025
Team ID	UNM1737WeatherWise
Project Name	AI WEATHERWISE
Maximum Marks	4 Marks

Solution Architecture:

AI WeatherWise uses a client-server architecture connecting the React frontend, Express backend, MongoDB database, OpenWeather API, and Gemini AI service. The architecture supports weather information, forecasts, user authentication, favorite locations, and AI-generated weather insights.

Main components:

- React.js frontend – provides the responsive user interface.
- Node.js / Express.js backend – handles routes, controllers, authentication, and application logic.
- MongoDB – stores user accounts and favorite locations.
- OpenWeather API – provides current weather and 5-day forecast data.
- Gemini AI – generates weather summaries and recommendations.

Architecture and Data Flow:

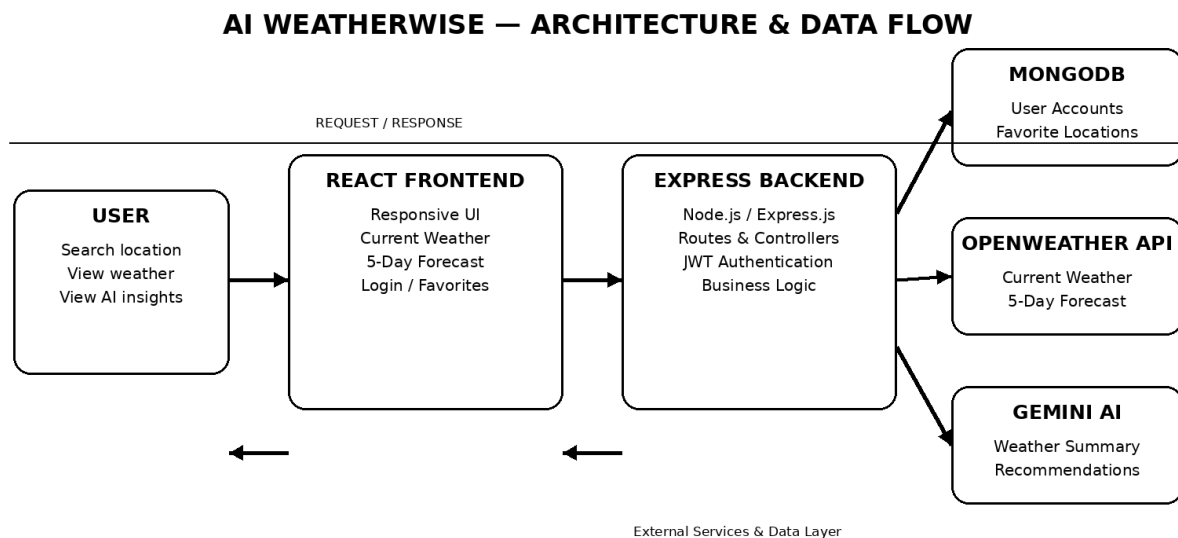


Figure 1: Architecture and data flow of the AI WeatherWise application

Data flow: User → React Frontend → Express Backend → MongoDB / OpenWeather API / Gemini AI → Express Backend → React Frontend → User.